COMMUTATIVITY OF THE SQUARE COMPARING PUSHFORWARD, DERIVED PUSHFORWARD, PULLBACK AND DERIVED PULLBACK

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the representatives are chosen explicitly and the comparison of the two paths is carried out with them.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Derived pullback, derived pushforward and their comparison maps with the naive functors. Same chapter, the constructions Lf^* and Rf_* together with the canonical maps f^* -> Lf^* and f_* -> Rf_*. Supplies the four vertices and the arrows of the displayed square.
- K-flat and K-injective resolutions. Same chapter, the existence of K-flat resolutions with flat terms and of K-injective replacements, by which the derived functors in the square are computed. Supplies the replacements with which both paths around the square are computed in the proof.

Source reference. Original lemma 06YS: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3207–3207. Navigation pointer invoked at proof line 4151; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 06YY: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3585–3585. Navigation pointer invoked at proof line 4151; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0G7E: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3508–3508. Navigation pointer invoked at proof line 4152; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $f : (Sh(\mathcal{C}), \mathcal{O}_\mathcal{C}) \to (Sh(\mathcal{D}), \mathcal{O}_\mathcal{D})$ be a morphism of ringed topoi. Let $\mathcal{K}^\bullet$ be a complex of $\mathcal{O}_\mathcal{C}$-modules. The diagram*

$$\begin{array}{ccc} Lf^*f_*\mathcal{K}^\bullet & \longrightarrow & f^*f_*\mathcal{K}^\bullet \\ \downarrow & & \downarrow \\ Lf^*Rf_*\mathcal{K}^\bullet & \longrightarrow & \mathcal{K}^\bullet \end{array}$$

coming from $Lf^* \to f^*$ on complexes, $f_* \to Rf_*$ on complexes, and adjunction $Lf^* \circ Rf_* \to id$ commutes in $D(\mathcal{O}_\mathcal{C})$.

Proof. We will use the existence of K-flat resolutions and K-injective resolutions, see Lemmas 06YS, 06YY, and 0G7E and the discussion above. Choose a quasi-isomorphism $\mathcal{K}^\bullet \to \mathcal{I}^\bullet$ where $\mathcal{I}^\bullet$ is K-injective as a complex of $\mathcal{O}_\mathcal{C}$-modules. Choose a quasi-isomorphism $\mathcal{Q}^\bullet \to f_*\mathcal{I}^\bullet$ where $\mathcal{Q}^\bullet$ is a K-flat complex of $\mathcal{O}_\mathcal{D}$-modules with flat terms. We can choose a K-flat complex of $\mathcal{O}_\mathcal{D}$-modules $\mathcal{P}^\bullet$ with flat terms and a diagram of morphisms of complexes

$$\begin{array}{ccc} \mathcal{P}^\bullet & \longrightarrow & f_*\mathcal{K}^\bullet \\ \downarrow & & \downarrow \\ \mathcal{Q}^\bullet & \longrightarrow & f_*\mathcal{I}^\bullet \end{array}$$

commutative up to homotopy where the top horizontal arrow is a quasi-isomorphism. Namely, we can first choose such a diagram for some complex $\mathcal{P}^\bullet$ because the quasi-isomorphisms form a multiplicative system in the homotopy category of complexes and then we can choose a resolution of $\mathcal{P}^\bullet$ by a K-flat complex with flat terms. Taking pullbacks we obtain a diagram of morphisms of complexes

$$\begin{array}{ccccc} f^*\mathcal{P}^\bullet & \longrightarrow & f^*f_*\mathcal{K}^\bullet & \longrightarrow & \mathcal{K}^\bullet \\ \downarrow & & \downarrow & & \downarrow \\ f^*\mathcal{Q}^\bullet & \longrightarrow & f^*f_*\mathcal{I}^\bullet & \longrightarrow & \mathcal{I}^\bullet \end{array}$$

commutative up to homotopy. The outer rectangle witnesses the truth of the statement in the lemma. □